

Vaibhav Kandpal

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EDUCATION

IIT Roorkee

Professional Certificate in Data Science & AI (6-month intensive)

Feb 2026 – Present

Remote

Dr. Akhilesh Das Gupta Institute of Professional Studies (GGSIPU)

Bachelor of Technology in Computer Science and Engineering

Aug 2024 – May 2028 (Expected)

New Delhi, India

AWARDS & ACHIEVEMENTS

- NASA Space Apps Challenge 2024: Top 5 in India (823 competing teams); built OrbitAir real-time air quality forecasting system integrating TEMPO satellite, EPA, OpenAQ & NOAA feeds. github.com/Vaibhav20k/ORBIT-AIR
- Paytm agent{a}thon winner - first place
- Deloitte Australia Data Analytics Job Simulation Forage certified (Jul 2025)

PROJECTS

DualSentry: Fraud & Anomaly Detection System | Go, Python, FastAPI, PostgreSQL, Apache Kafka, Docker, XGBoost, gRPC, Redis

<https://github.com/Vaibhav20k/DualSentry-model.git>

- Architected an event-driven fraud detection platform using Go microservices, gRPC, Apache Kafka, PostgreSQL, and Docker, processing transactions through a real-time pipeline with feature engineering, ML inference, and fraud decisioning (ALLOW / REVIEW / BLOCK) while persisting predictions for downstream analytics.
- Built an ML anomaly detection engine using XGBoost and Isolation Forest with 20+ engineered behavioral features including transaction velocity, amount deviation, merchant frequency, device history, location risk, and active-hour analysis; implemented automated CSV dataset generation for continuous model retraining and evaluation.
- Developed an end-to-end fraud intelligence workflow integrating feature extraction, baseline profiling, model inference, decision engine, and prediction persistence, achieving 100% successful end-to-end transaction processing from gRPC ingestion through Kafka streaming to PostgreSQL-backed fraud prediction storage with versioned model metadata.

OrbitAir: AI-Based Air Quality Forecasting System | Python, React, TimescaleDB, FastAPI, Docker

<https://github.com/Vaibhav20k/ORBIT-AIR>

- Engineered multi-source ETL pipeline fusing TEMPO satellite, EPA, OpenAQ & NOAA feeds into TimescaleDB via async batch processing reduced pipeline runtime from 8 min to under 2 min (4× throughput gain).
- Delivered Random Forest + SHAP explainability pipeline achieving 98% prediction accuracy; surfaced top 5 regional pollution drivers on a React + Leaflet dashboard.
- Refactored React component tree with lazy-loaded map tiles; improved Lighthouse performance score from 61 → 89. Built and deployed end-to-end in 48 hours for NASA Space Apps Challenge.

Sahai: AI Mental Health & Lifestyle Ecosystem | FastAPI, Django, PostgreSQL, Redis, FAISS, Docker, WebSockets

https://github.com/Vaibhav20k/Sahai_Project

- Architected a RAG pipeline with FAISS vector search over curated mental health documentation; integrated LoRA fine-tuned LLM (OpenAI / Hugging Face) for empathetic, context-aware, and safe conversational responses with crisis detection guardrails.
- Built real-time WebSocket chat with JWT authentication and role-based access control (students, counselors, admins); deployed with Docker containerization, NGINX reverse proxy, and GitHub Actions CI/CD.
- Designed mood tracking with PHQ-9 standardized assessments, peer support forums, college-specific circles, and counselor booking system with automated crisis escalation routing to emergency contacts.

EXPERIENCE

Cimplifie | Full sack intern

Jun 2026 – July 2026 (In-office)

- Shipped 4 client-facing React dashboards consuming internal REST APIs, reducing average client reporting turnaround from 3 days to same-day by replacing manual Excel exports with live data views.
- Integrated GitHub Actions CI/CD pipeline for the company's Node.js backend automated lint, test, and Vercel preview deploy on every PR, cutting QA review time by 50%.
- Wrote browser-automation scripts in Python to validate UI flows across staging environments after each release, catching 11 regression bugs before they reached production.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++

Frontend: React.js, Next.js, HTML5/CSS3, Tailwind CSS, shadcn/ui

Backend: FastAPI, Node.js, Express.js, Django, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, SQLite, TimescaleDB, Redis

ML: Scikit-learn, NumPy, Pandas, Random Forest, SHAP, FAISS (RAG), Model Evaluation, Feature Engineering, Hyperparameter Tuning, Model Monitoring, A/B Testing

DevOps & Tools: Docker, Git, GitHub Actions CI/CD, NGINX, Vercel, Selenium, Microservices, API Design

Data Science: ETL, Data Pipeline, Distributed Systems, TimescaleDB, Data Warehouse, DuckDB, Arrow, Parquet